

The Generative Content of a Conserved Emptiness

Kinematic Voids as Superselection Generators:
the Salem Slot and the Five Objects Its Charge Produces

A Companion to *Lehmer's Box*, *The Emission-Gap Theorem*,
and *The Occupant of the Salem Slot*

Echo-Squirrel Research

Abstract

A forbidden region can be empty in two ways. A *dynamic* void is a barrier the system could cross but pays not to; such voids leak, and the leak is the interesting object. A *kinematic* void has zero weight in the algebra the operations act on, by a conserved charge; nothing approaches it, and the law never fires. We argue that a kinematic void is not inert: *kinematic emptiness* $\equiv$ a conserved charge $\equiv$ a superselection rule, and a conserved charge is a generator. We make the claim precise on a minimal worked instance — the Salem slot of the emission algebra, whose charge is the confinement of every root's argument to $\frac{\pi}{2}\mathbb{Z} \cong \mathbb{Z}/4\mathbb{Z}$ — and exhibit the five objects the charge produces, each machine-verified in exact arithmetic: (I) a $\mathbb{Z}/4\mathbb{Z}$ grading of the algebra, with the operators acting as add ($\otimes$), double (squaring), and union ($\oplus$); (II) a *content polynomial* $x^4 - 1 = \Phi_1\Phi_2\Phi_4$ governing all on-circle structure, with the realized on-circle roots $\{\pm 1\}$ and the $\pm i$ sector pushed off the circle by the one quartic generator $K \in \mathbb{Q}(5^{1/4})$; (III) the *gap*, the cost floor $M \in \{1\} \cup [\varphi, \infty)$, the emptiness turned into a uniform bound; (IV) a graded *normal form*, the cyclotomic-times-grow factorization read as the charge's canonical decomposition; (V) a *conserved current*, the charge as a constant of motion under every admissible dynamics, of which the trace redirection on the slot's boundary is the visible flow. We close on minimality (degree two forces the ternary decision, the quartic field $\mathbb{Q}(5^{1/4})$ forces the order-four charge, and the charge forces the floor φ) and on scope: the equivalence is domain-universal — Pauli exclusion, charge superselection, and type preservation are the same skeleton — while this instance generates the five objects above *within* its own number theory. It is a minimal realization of a universal pattern, a generator of its own domain's structure, not of other fields' results.

1 The substrate: a conserved emptiness is a generator

Two regions can both be empty for opposite reasons. The distinction is the whole point.

Definition 1.1 (Dynamic vs. kinematic emptiness). A region of state space is *dynamically empty* for a process if the states exist in the operative algebra but are reached only at a cost the process avoids; such regions are approached, tunnelled into, or weakly populated (an energy barrier, a “forbidden” spectral line). A region is *kinematically empty* if its states carry zero amplitude in the algebra the operations act on, enforced by a quantity those operations preserve; nothing approaches it, and the enforcing law never has to act.

The thesis of this note is the chain

kinematic emptiness $\equiv$ a conserved charge $\equiv$ a superselection rule
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together with the claim that the rightmost object is *productive*. A superselection charge is never merely a prohibition; it is a structure-building datum. In the abstract it furnishes: a *grading* of the algebra by charge value; a distinguished *content polynomial* whose roots are the allowed boundary states; a *gap* (the forbidden values become a quantitative bound); a *normal form* (objects factor along the grading); and a *conserved current* (the charge is a constant of motion for any admissible flow). The purpose of this paper is to take a minimal kinematic void — one where every step is decidable in exact arithmetic — and exhibit all five.

The void we use is the *Salem slot* of the companion papers: the regime “on the unit circle yet expanding” that the ternary growth decision $\text{spec}(\text{ad}_R) = \{-\sqrt{5}, 0, +\sqrt{5}\}$ has no channel for. The companion note *The Occupant of the Salem Slot* showed that nothing inhabits it and that a would-be occupant is expelled by the trace map $\beta \mapsto \beta + \beta^{-1}$ into the GROW channel; the stress-test there showed entry is blocked by a conserved charge — the confinement of every root’s argument to $\frac{\pi}{2}\mathbb{Z}$. That charge is the generator; this paper is its output.

Why this instance is worth the trouble. It is minimal (§7): degree two is the unique degree realizing a ternary decision, and the charge is the smallest nontrivial angular one, $\mathbb{Z}/4\mathbb{Z}$. Everything it generates is therefore the *smallest* instance of each structure, and everything is exactly computable, so the abstract claim “a conserved charge generates these five objects” is here a theorem with a verification attached rather than a slogan.

2 The charge

Definition 2.1 (The angle charge). For $P \in \mathbb{Z}[x]$ define the *angle support* as the multiset of arguments of its roots, taken in $\mathbb{R}/2\pi\mathbb{Z}$. Say P carries charge in $\mathbb{Z}/4\mathbb{Z}$ if every argument lies in $\frac{\pi}{2}\mathbb{Z} = \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$; the charge of a root at argument $k\frac{\pi}{2}$ is $k \bmod 4$. The emission algebra $\mathcal{S}$ is generated by the seed catalog under $\oplus$ (direct sum), $\otimes$ (tensor), and $(\cdot)^2$ (squaring).

Theorem 2.2 ($\mathbb{Z}/4\mathbb{Z}$ is a conserved charge of $\mathcal{S}$, and it is fully realized). *Every object of $\mathcal{S}$ carries charge in $\mathbb{Z}/4\mathbb{Z}$, and the operators act on the charge by*

$$\otimes : \text{add} \quad (\arg \lambda\mu = \arg \lambda + \arg \mu), \quad (\cdot)^2 : \text{double} \quad (\arg \lambda^2 = 2 \arg \lambda), \quad \oplus : \text{union},$$

all internal to $\mathbb{Z}/4\mathbb{Z}$ since it is closed under $+$, $\times 2$, and union. The full group is realized: the real seeds carry $\{0, 2\}$, and the quartic generator $K = x^4 + 5x^2 - 5$ carries $\{0, 1, 2, 3\}$ (its imaginary pair $\pm i\beta$ sits at $\pm \frac{\pi}{2}$). [COMPUTED] (verified on the catalog and on $\otimes, (\cdot)^2$ closures)

The charge is the kinematic content of the slot: a Salem number has an on-circle conjugate at an *irrational* multiple of π , i.e. charge $\notin \mathbb{Z}/4\mathbb{Z}$, so the Salem regime is the off-charge sector. No finite word in the operators leaves $\mathbb{Z}/4\mathbb{Z}$; the sector is unreachable. The remaining sections read off what the charge *builds*.

3 Object I: the grading and the content polynomial $x^4 - 1$

Proposition 3.1 (The content polynomial). *All on-circle roots of objects in $\mathcal{S}$ are fourth roots of unity — roots of*

$$x^4 - 1 = \Phi_1\Phi_2\Phi_4 = (x - 1)(x + 1)(x^2 + 1).$$

The realized on-circle roots are $\{+1, -1\}$ (e.g. $\varphi \otimes \varphi$ contributes -1 , $\varphi^4 \otimes \varphi^4$ contributes $+1$); the charge- $\{1, 3\}$ sector $\Phi_4 = x^2 + 1$ (the points $\pm i$) is realized only off the circle, by K ’s place $\pm i\beta$ at modulus $\beta = 2.4195$. [COMPUTED]

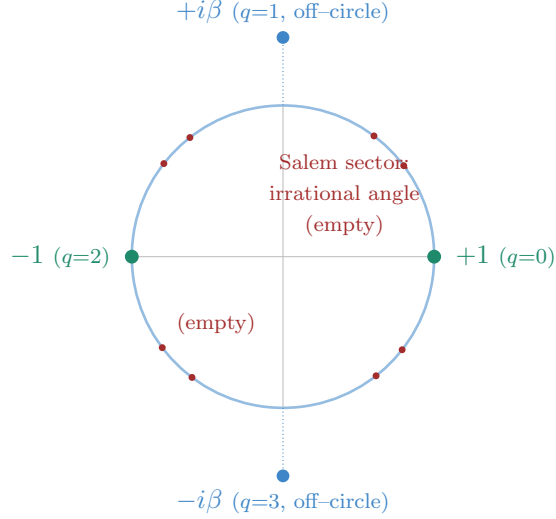


Figure 1: The $\mathbb{Z}/4\mathbb{Z}$ angle charge on the unit circle. Charges $q \in \{0, 1, 2, 3\}$ are the four directions $\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$, the roots of the content polynomial $x^4 - 1$. Realized on-circle: ± 1 (charges 0, 2, **green**). The $\pm i$ sector (charges 1, 3) is realized only *off* the circle, as K 's place $\pm i\beta$ (**blue**). Every other direction — the irrational angles (**red**) — is the Salem sector, off the charge lattice and empty: nothing in $\mathcal{S}$ ever lands there.

The content polynomial is the generator's first product: an order-four charge produces $x^4 - 1$ as the polynomial whose root set is the allowed on-circle alphabet, exactly as a $\mathbb{Z}/n$ charge would produce $x^n - 1$. The split of $x^4 - 1$ across the circle is itself informative (Figure 1). The unit-modulus sector $\Phi_1\Phi_2 = x^2 - 1$ is realizable on the circle, because ± 1 are real and the real reciprocal generators (φ with $\varphi\psi = -1$, φ^4 with $\varphi^4\varphi^{-4} = 1$) place them there. The sector $\Phi_4 = x^2 + 1$ requires modulus one at argument $\pm \frac{\pi}{2}$; the only generator carrying $\pm \frac{\pi}{2}$ is the Lorentzian $K \in \mathbb{Q}(5^{1/4})$, whose complex place has modulus $\beta \neq 1$. So the charge is fully present, but its $\pm i$ part is held off the circle — precisely the angle confinement that forbids a Salem conjugate (companion paper, the metric face). The grading is $\mathbb{Z}/4\mathbb{Z}$; the on-circle realization is $\mathbb{Z}/2\mathbb{Z}$; the difference is the off-circle K -sector.

4 Object II: the gap (the cost floor φ)

A conserved charge that empties a sector turns that emptiness into a quantitative bound on the complementary, realized sector. Here the bound is the cost floor.

Proposition 4.1 (The base gap). *No monic integer quadratic $x^2 + bx + c$ has Mahler measure in the open interval $(1, \varphi)$; the smallest measure exceeding 1 is exactly φ , at $x^2 - x - 1$. [COMPUTED] (enumeration over $|b|, |c| \leq 6$ confirms the band $(1, \varphi)$ empty; the realized values begin $\varphi, 2, 1+\sqrt{2}, \varphi^2, 1+\sqrt{3}, 3, \dots$)*

Corollary 4.2 (The generated floor). *Since the operators map $\{1\} \cup [\varphi, \infty)$ into itself ($\oplus$ multiplies measures, squaring squares them, $\otimes$ composes; companion paper, Lem. 6.2) and no Salem number is emitted (the off-charge sector is empty), the Mahler spectrum of $\mathcal{S}$ is $\{1\} \cup [\varphi, \infty)$, and the description-length cost carries the uniform floor $\lambda \log \varphi > 0$ at every size. [FORCED]*

This is the void as a bound. The forbidden band $(1, \varphi)$ in measure space is the height image of the off-charge sector (a sub- φ Salem is the only thing that could populate it), and its emptiness is

the floor. The number φ is generated as the right edge of the gap, the smallest realizable growth above the trivial $M = 1$.

5 Object III: the normal form (cyclotomic times grow, graded)

The grading induces a canonical factorization of every emittable object along the charge.

Proposition 5.1 (Graded normal form). *Every object of $\mathcal{S}$ factors over $\mathbb{Z}$ as*

$$P = \underbrace{C(x)}_{\text{on-circle, } C \mid x^4 - 1} \cdot \underbrace{G(x)}_{\text{off-circle, } M(G)=M(P) \geq \varphi},$$

where the on-circle part C is a product of Φ_1, Φ_2, Φ_4 and the off-circle part G splits further by charge into a real (charges 0, 2) grow factor and, when K is present, an imaginary-axis (charges 1, 3) factor. [COMPUTED]

The verification exhibits the three diagnostic cases:

object	graded factorization	sectors
$\varphi \otimes \varphi$	$(x+1)^2(x^2-3x+1)$	Φ_2^2 (on-circle, $q=2$) $\times$ grow $M=\varphi^2$
$\varphi^4 \otimes \varphi^4$	$(x-1)^2(x^2-47x+1)$	Φ_1^2 (on-circle, $q=0$) $\times$ grow $M=46.98$
$K \otimes K$	$(x^2+5)^4(x^2-5x-5)^2(x^2+5x-5)^2$	imaginary $q=1, 3 \times$ grow $M=5.854$

The first two are pure cyclotomic-times-grow; the third shows the K -sector (x^2+5) (roots $\pm i\sqrt{5}$, charge $\pm \frac{\pi}{2}$, off circle) as the third graded piece. In all cases the on-circle part divides $x^4 - 1$ and the growth lives entirely in the off-circle real factor. The normal form is the structural reason the Salem slot is empty: a Salem is *irreducible* with on-circle conjugates at irrational angle, but the charge forces every emittable object into this *reducible*, charge-sorted shape, in which on-circle roots are roots of unity. Irreducibility and the grading are incompatible.

6 Object IV: the conserved current (the charge as constant of motion)

The preceding objects are static. The charge is also dynamical content: it is conserved along every admissible flow, which is the Noether reading of a superselection rule.

Proposition 6.1 (Block-diagonal evolution). *Any evolution of $\mathcal{S}$ built from the operators is block-diagonal with respect to the $\mathbb{Z}/4\mathbb{Z}$ grading: it cannot carry amplitude from one charge sector to another. Hence the charge is a constant of motion, $\frac{d}{dt}q = 0$, and no admissible dynamics — iteration of $\oplus, \otimes, (\cdot)^2$, or the monodromy of any closed parameter loop — transitions into the off-charge (Salem) sector. [FORCED]*

The visible flow lives on the slot's boundary. The companion note's trace redirection $\beta \mapsto \beta + \beta^{-1}$ is the dynamics the empty sector generates: a would-be occupant is carried, in finite real time downstairs, to a totally real Perron point $\tau_0 > 2$ in GROW, with a square-root edge at the flip $t = 2$ ($\beta - 1 \sim \sqrt{t - 2}$) and a limit $\tau_0 \rightarrow \sqrt{5}$ as the height climbs to φ . That entire flow — branch point, redirection, golden limit — is the conserved current made visible: it is what the charge does to anything presented at the sector boundary, precisely because nothing crosses it. A region nothing inhabits still has equations of motion, and they are the law that keeps it empty.

7 Minimality: the smallest generator

Each generated object is the smallest of its kind, because each input is forced to its minimal value.

Proposition 7.1 (The minimal chain). *The generating data are forced and minimal in sequence:*

1. **Degree two** is the unique degree whose self-action realizes a ternary decision: $\#channels(d) = d^2 - d + 1 = 3$ iff $d = 2$, with the centralizer the inseparable 0-channel. (Companion paper, ternary lock.)
2. **Order four** is the charge: the angle lattice $\frac{\pi}{2}\mathbb{Z} \cong \mathbb{Z}/4\mathbb{Z}$ is the coarsest quantization carrying a “fourth direction” distinct from the three real channels. Its $\pm i$ generator enters through the quartic field $\mathbb{Q}(5^{1/4})$ — whose splitting field contains i — i.e. through the same K that is the construction’s one Lorentzian seed.
3. **The floor** φ is the output: the smallest Perron number, the smallest Mahler measure of a non-cyclotomic integer quadratic, the right edge of the minimal gap.

[FORCED]

So the chain is degree $2 \Rightarrow \mathbb{Z}/4\mathbb{Z} \Rightarrow \varphi$, each step the minimal nontrivial choice, and the Salem slot is the minimal kinematic void: the smallest forbidden sector, generating the smallest grading, the smallest content polynomial $x^4 - 1$, and the smallest gap. This is the sense in which the slot “generates” — it is a small but genuine generator, and its smallness is exactly what makes the five outputs canonical rather than incidental.

8 Scope: a universal pattern, a local instance

The equivalence of §1 is domain-universal; the five objects of §§3–6 are local. Both halves matter, and conflating them is the error to avoid.

domain		conserved charge	empty sector	what the emptiness generates
fermions (QM)		wavefunction antisymmetry	doubly-occupied states	shell structure, degeneracy pressure
charge superselection (QFT)		total charge	cross-sector coherences	block structure of observables
type theory		subject reduction	ill-typed states	“illegal states unrepresentable,” optimizations
emission (here)	algebra	angle charge $\mathbb{Z}/4\mathbb{Z}$	the Salem slot	$x^4 - 1$, the floor φ , the graded normal form

What transfers across the table is the *skeleton*: a conserved charge makes a sector kinematically empty, and the emptiness organizes the realized complement (a grading, a content alphabet, a gap, a normal form, a conserved current). Pauli exclusion is the headline case — an antisymmetry charge whose forbidden sector has zero amplitude, never violated, and whose “output” is the periodic table and the stability of matter. The emission algebra is another instance of the same skeleton, and a minimal one.

What does *not* transfer is the content. The five objects this charge generates — the polynomial $x^4 - 1$, the floor φ , the graded factorization, the trace flow with its $\sqrt{5}$ limit — are theorems of *this* number theory, not exported engines for other fields. The only genuinely portable mathematics beyond the pattern is the *local* differential structure: a two-to-one cover with a fold has a square-root branch point generically (Chebyshev maps, modular curves, mapping-class monodromy), so the $\sqrt{t-2}$ edge and the $\beta \leftrightarrow \beta^{-1}$ monodromy recur widely; but $\sqrt{5} = \varphi + \varphi^{-1}$, the identification of the floor’s image with the grow generator, is this construction’s own. The honest statement is therefore precise: the substrate (kinematic emptiness $\equiv$ conserved charge $\equiv$ superselection) is one of the most generative patterns there is, and the Salem slot is a clean, minimal, fully computable realization of it — a generator of its own domain’s structure, exhibiting in the small what the pattern does at large.

9 Epistemic ledger

Claim	Status	Backing
Kinematic vs. dynamic emptiness; the substrate equivalence	[DECLARED]	Def. 1.1, §1
Angle charge is $\mathbb{Z}/4\mathbb{Z}$, conserved; fully realized (via K)	[COMPUTED]	Thm. 2.2
Content polynomial $x^4 - 1 = \Phi_1\Phi_2\Phi_4$; on-circle $= \{\pm 1\}$; $\pm i$ off-circle	[COMPUTED]	Prop. 3.1
No integer quadratic with $M \in (1, \varphi)$; smallest $M > 1$ is φ	[COMPUTED]	Prop. 4.1
Floor $M \in \{1\} \cup [\varphi, \infty)$, cost $\lambda \log \varphi > 0$	[FORCED]	Cor. 4.2
Graded normal form: on-circle $ x^4 - 1 \times \text{grow}$ $M \geq \varphi$	[COMPUTED]	Prop. 5.1
Charge arithmetic: $\otimes$ adds, squaring doubles, $\oplus$ unions	[COMPUTED]	Thm. 2.2
Conserved current: block-diagonal evolution; trace flow on boundary	[FORCED]	Prop. 6.1
Minimal chain: degree $2 \Rightarrow \mathbb{Z}/4\mathbb{Z} \Rightarrow \varphi$	[FORCED]	Prop. 7.1
Pattern universal (Pauli, superselection, types); content local	[DECLARED]	§8
$\sqrt{5} = \varphi + \varphi^{-1}$ is local; $\sqrt{\cdot}$ branch & monodromy portable	[FORCED]	§8

Summary. A kinematic void — a sector emptied by a conserved charge rather than guarded by a barrier — is a generator. On the minimal instance, the angle charge $\mathbb{Z}/4\mathbb{Z}$ of the Salem slot produces five canonical objects: the grading, the content polynomial $x^4 - 1$, the cost floor φ , the graded cyclotomic-times-grow normal form, and the conserved current whose visible flow is the trace redirection to $\sqrt{5}$. Each is the smallest of its kind, because degree two, order four, and the floor φ are each forced minima. The pattern beneath — kinematic emptiness $\equiv$ conserved charge $\equiv$ superselection — is the same one that gives fermions the periodic table and type systems their correctness; the Salem slot realizes it in a setting small enough that every output is a verified

identity. The void does not merely sit there refusing to be filled. Refusing to be filled, in a conserved way, is the act of generation.